

PROJECT REPORT

Implement Client Script & UI Policy (Incident)

Team ID: SWTID-2026-1012

Team Size: 3

Team Leader: Raman P

Team Members: Sivasakthi P, Selvakannan C

1. Introduction

1.1 Project Overview

This project implements a UI Policy named 'High Impact Control' along with three Client Scripts (onChange, onSubmit, and onCellEdit) on the Incident table in ServiceNow. The objective is to make Assignment group mandatory and Urgency read-only (and auto-set) when Impact is High, enforce Assigned To as mandatory for High Impact incidents on submit, and prevent the State field from being changed directly through list-view inline editing.

1.2 Purpose

The purpose of this solution is to ensure every High Impact Incident is properly owned and correctly flagged, while protecting the integrity of the State field from uncontrolled list-view edits, thereby improving data quality, accountability, and incident resolution speed.

2. Ideation Phase

2.1 Problem Statement

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	A ServiceNow Incident support agent	make sure every High Impact incident has an owner and correct Urgency	I sometimes save High Impact incidents without an Assigned To or with the wrong Urgency	there's no automated validation enforcing Assigned To and Urgency when Impact = High	Frustrated and worried about unassigned critical incidents
PS-2	A ServiceNow admin/team lead	protect the integrity of the State field on Incident records	agents sometimes change State directly from the list view without proper review	there's no restriction preventing inline State edits from the list	Concerned about uncontrolled status changes

2.2 Empathy Map Canvas

Section	Details
Think & Feel	Worried about missing Assigned To on High Impact incidents; frustrated by escalations without an owner
See	High Impact incidents unassigned; Urgency inconsistently set; State changed carelessly from list view
Hear	Manager insists High Impact incidents must have an owner and correct Urgency
Say & Do	Sets Impact quickly without always assigning an owner; edits State directly from list view sometimes

Pain	High Impact incidents saved without an assignee; State changed accidentally from list
Gain	Wants Assigned To enforced for High Impact; Urgency auto-updated; State changed only via form

2.3 Brainstorming

How might we ensure High Impact Incidents always have a mandatory owner and correct Urgency, while protecting the State field from uncontrolled list-view edits?

- Create UI Policy 'High Impact Control' to make Urgency read-only when Impact = High
- Add onChange Client Script to auto-set Urgency to High when Impact is High
- Add onSubmit Client Script to block save when Assigned To is empty for High Impact
- Add onCellEdit Client Script to block direct State field edits from the list view

Final Selected Approach: UI Policy 'High Impact Control' + three Client Scripts (onChange, onSubmit, onCellEdit).

3. Requirement Analysis

3.1 Customer Journey Map

Scenario: Agent creates and updates a High Impact Incident record. Steps flow from login -> setting Impact=High (Urgency auto-sets, becomes read-only) -> submitting (Assigned To checked) -> list view (State edit blocked).

3.2 Solution Requirement

Functional Requirements:

FR No.	Requirement (Epic)	Sub Requirement
FR-1	UI Policy - High Impact Control	Condition: Impact = 1-High / Active = true / Reverse if false = true
FR-2	UI Policy Action - Urgency	Urgency field Read-only when Impact = High
FR-3	Client Script - onChange	Auto-set Urgency to High when Impact changes to High
FR-4	Client Script - onSubmit	Block save if Assigned To empty when Impact = High
FR-5	Client Script - onCellEdit	Block State field edit from Incident list view

Non-Functional Requirements:

NFR No.	Requirement	Description
NFR-1	Usability	Error/alert messages must be clear and actionable
NFR-2	Security	Only ITIL/incident manager role can modify configuration
NFR-3	Reliability	Validation consistent across change, submit, and list edit
NFR-4	Performance	Scripts execute in real-time (<1 sec)
NFR-5	Availability	Works across sessions without cache refresh
NFR-6	Scalability	Pattern extends to other conditions/tables

3.3 Data Flow Diagram

Agent -> Incident Form -> High Impact Control UI Policy -> Client Script Engine (onChange/onSubmit) -> Incident Table (DB) -> List View -> onCellEdit Validation (State) -> Updated Record

3.4 Technology Stack

S.No	Component	Technology
1	User Interface	ServiceNow UI (Now Platform)
2	Declarative Logic	UI Policy 'High Impact Control' + UI Policy Action

3	Client-side Scripting	Client Scripts (onChange, onSubmit, onCellEdit), GlideForm (g_form) API
4	Database	ServiceNow native database
5	Platform	ServiceNow Cloud (PaaS)
6	Access Control	ServiceNow ACLs

4. Project Design

4.1 Problem Solution Fit

Customer Segment: ServiceNow Incident agents/admins handling High Impact incidents. Root Cause: no dynamic validation enforcing Assigned To for High Impact or protecting State from list edits. Solution: UI Policy 'High Impact Control' + onChange/onSubmit/onCellEdit Client Scripts.

4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement	High Impact Incidents saved without Assigned To; Urgency not auto-aligned; State changed from list view without review
2	Idea / Solution	UI Policy makes Urgency read-only when Impact=High; onChange auto-sets Urgency; onSubmit enforces Assigned To; onCellEdit blocks State list edits
3	Novelty	Combines declarative UI Policy with three Client Script types covering auto-population, mandatory validation, and list-view protection
4	Social Impact	Ensures every High Impact incident is owned and correctly flagged; protects data integrity
5	Business Model	Internal process-improvement configuration; value is operational efficiency
6	Scalability	Extends to other Impact/Urgency combinations or tables

4.3 Solution Architecture

Agent Browser -> Incident Form/List View -> High Impact Control UI Policy Engine (Impact=High) -> UI Policy Action (Urgency Read-only) -> onChange (auto-set Urgency) -> onSubmit (Assigned To check) -> onCellEdit (State block) -> ServiceNow Server (ACL Layer) -> Incident Table (Database).

5. Project Planning & Scheduling

5.1 Project Planning

Sprint	Epic	Story No.	User Story / Task	Points	Priority
Sprint-1	UI Policy - High Impact Control	USN-1	Create UI Policy with condition Impact = High	2	High
Sprint-1	UI Policy - High Impact Control	USN-2	Enable Reverse if False	1	High
Sprint-1	UI Policy Action	USN-3	Add action to make Urgency read-only	2	High
Sprint-1	Client Script - onChange	USN-4	Auto-set Urgency to High on Impact change	3	High
Sprint-2	Client Script - onSubmit	USN-5	Block save if Assigned To empty (High Impact)	3	High
Sprint-2	Client Script - onCellEdit	USN-6	Block State edit from list view	3	Medium
Sprint-2	Testing	USN-7 to USN-11	5 testing activities (Milestone 6)	7	High/Medium

Story Points: Sprint 1 = 8, Sprint 2 = 13, Total = 21. Velocity = 21/2 = 10.5 per Sprint.

6. Functional and Performance Testing

6.1 Performance Testing

Test Case	Scenario	Expected Result	Result
FT-01	Test Mandatory Enforcement (Activity 1)	Save blocked, error shown when Assigned To empty on High Impact	Pass
FT-02	Test Successful Save (Activity 2)	Saves successfully once Assigned To filled	Pass
FT-03	Reverse Condition Test (Activity 3)	Assigned To not mandatory, Urgency editable when Impact != High	Pass
FT-04	Test List Edit Blocking (Activity 4)	Alert shown, State unchanged on list-view edit attempt	Pass
FT-05	Test Form-Based Update (Activity 5)	State change via form saves successfully	Pass
PT-01	Client Script Response Time	Updates in under 1 second	Pass

7. Results

7.1 Output Screenshots

The following screenshots show the Client Scripts and the results of the testing activities in ServiceNow.

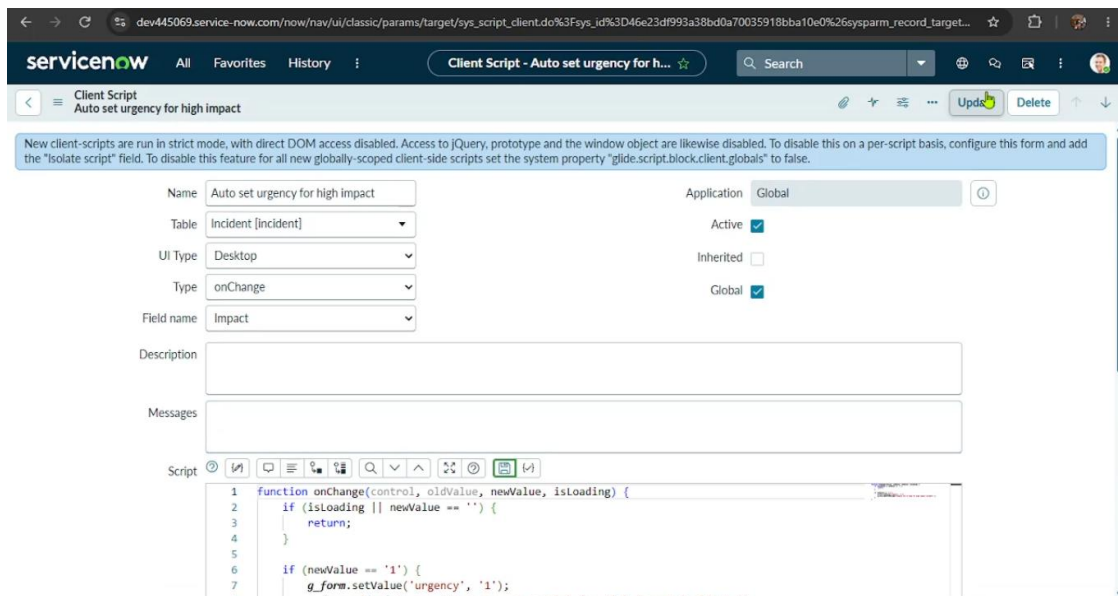


Figure 1: onChange Client Script 'Auto set urgency for high impact' (Field: Impact)

servicenow All Favorites History Client Script - Prevent save if Assign...

Client Script
Prevent save if Assigned To missing

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Prevent save if Assigned To missing Application: Global

Table: Incident [incident] Active: ☒

UI Type: Desktop Inherited: ☐

Type: onSubmit Global: ☒

Description:

Messages:

Script

```

1 function onSubmit() {
2   if (g_form.getValue('impact') == '1' &&
3       g_form.getValue('assigned_to') == '') {
4
5     g_form.showErrorMessage(
6       'assigned_to',
7       'Assigned To is mandatory for High impact incidents.'
8     );
9   }
10 }

```

Figure 2: onSubmit Client Script 'Prevent save if Assigned To missing'

servicenow All Favorites History Admin Client Script - New Record

Client Script
New record

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Prevent state change via list edit Application: Global

Table: Incident [incident] Active: ☒

UI Type: Desktop Inherited: ☐

Type: onCellEdit Global: ☒

Field name: State

Description:

Messages:

Script

```

1 function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
2
3   alert('State cannot be updated using list editing. Please open the Incident.');
```

Save

57:01 1:06:21 90% 57:01/1:06:21

Figure 3: onCellEdit Client Script 'Prevent state change via list edit' (Field: State)

dev445069.service-now.com/now/nav/ui/classic/params/target/incident.do%3Fsys_id%3Dbd27fd3593e38bd0a70035918bba10e6%26sysparm_view%3D%26sysparm...

servicenow All Favorites History Admin Incident - INC0010004 Search

Incident INC0010004 Discuss Follow Update Resolve Delete Responses

Urgency set to High for High impact incident.

Number INC0010004 Channel -- None --

* Caller Abel Tuter State New

Category Inquiry / Help Impact 1 - High

Subcategory -- None -- Urgency 1 - High

Service Priority 1 - Critical

Service offering * Assignment group

Configuration item Assigned to

* Short description practice

Description

Related Search Results

Notes Related Records Resolution Information

Figure 4: Impact set to High - Urgency auto-set to High (read-only) with info message

servicenow All Favorites History Workspaces Admin Incident - Create INC0010012 Search

Incident New record Submit Resolve

Number INC0010012 Channel -- None --

* Caller Adela Cervantsz State New

Category Inquiry / Help Impact 1 - High

Subcategory -- None -- Urgency 1 - High

Service Priority 1 - Critical

Service offering * Assignment group Software

Configuration item Assigned to

* Short description testing UI policy

Description

Assigned To is mandatory for High impact incidents.

Related Search Results

Related Search Q testing UI policy Knowledge & Catalog (AI)

Commany policies No description

Whitelist IP Need to whitelist an IP address?

Figure 5: Activity 1 - Save blocked, error 'Assigned To is mandatory for High impact incidents'

servicenow All Incident

Incident - Create INC0010012

Submit Resolve

Number INC0010012

* Caller Adela Cervantsz

Category Inquiry / Help

Subcategory -- None --

Service

Service offering

Configuration item

* Short description testing UI policy

Description

Channel -- None --

State New

Impact 1 - High

Urgency 1 - High

Priority 1 - Critical

* Assignment group Software

Assigned to Beth Anglin

Related Search

Company policies No description

Whitelist IP Need to whitelist an IP address?

Firewall Rule Change Cisco Firewall Appliance

Figure 6: Activity 2 - Assigned To filled, Incident ready to submit successfully

servicenow All Incident INC0010012

Discuss Follow Update Resolve Delete

Number INC0010012

* Caller Adela Cervantsz

Category Inquiry / Help

Subcategory -- None --

Service

Service offering

Configuration item

* Short description testing UI policy

Description

Channel -- None --

State New

Impact 2 - Medium

Urgency 1 - High

Priority 2 - High

Assignment group Software

Assigned to

Country -- None --

States -- None --

Notes Related Records Resolution Information

Figure 7: Activity 3 - Impact changed to Medium, Urgency becomes editable (reverse condition)

servicenow All Incidents

dev354432.servicenow.com says

State cannot be changed from the list. Please open the incident.

OK

Category	Assignment group	Assigned to	Updated	Updated by
Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
Software	(empty)	(empty)	2018-12-12 23:18:55	admin
Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
Assessment : ATF Assessor	survey user	5 - Planning	2019-07-29 11:49:28	admin
ATF : Test1	System Administrator	5 - Planning	2019-07-22 14:05:48	admin
ATF:TEST2	survey user	5 - Planning	2021-01-21 15:31:42	admin
Need access to the common drive.	David Miller	4 - Low	2018-12-12 23:28:49	admin
Employee payroll application server is down.	David Miller	1 - Critical	2025-12-03 18:11:31	system
Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	2020-06-07 09:04:34	admin
Unable to access team	David Miller	3 -	2016-08-10	admin

1 to 20 of 41

Figure 8: Activity 4 - List view State edit blocked with alert message

8. Advantages & Disadvantages

Advantages

- Ensures every High Impact incident has an owner and correct Urgency
- Protects State field from careless or unauthorized list-view changes
- Real-time validation improves data quality with no custom UI page needed
- Reusable pattern for other Impact/Urgency combinations

Disadvantages

- Client Scripts run in the browser and can be bypassed via direct API/background scripts
- Requires ITIL/admin role knowledge to configure and maintain
- Multiple script types increase maintenance overhead if rules change

9. Conclusion

This project successfully demonstrates the implementation of UI Policies, UI Policy Actions, and Client Scripts on the ServiceNow Incident table. The configuration enables dynamic field behavior, validation, and controlled data entry based on user actions and predefined conditions. By testing different scenarios, the Incident form and list view become more user-friendly, consistent, and reliable, helping improve overall data quality and incident management efficiency.

10. Future Scope

- Extend the High Impact Control pattern to other Impact/Urgency combinations
- Add server-side Business Rules as a backup validation layer to prevent bypass via API
- Integrate with Flow Designer to notify managers of unassigned High Impact incidents
- Apply similar list-edit protection to other critical fields beyond State

11. Appendix

Source Code

onChange Client Script - Auto set urgency for high impact:

```
function onChange(control, oldValue, newValue, isLoading) { if (isLoading || newValue == "") { return; }
if (newValue == '1') { g_form.setValue('urgency', '1'); g_form.addInfoMessage('Urgency set to High for
High impact incident.');
```

onSubmit Client Script - Prevent save if Assigned To missing:

```
function onSubmit() { if (g_form.getValue('impact') == '1' && g_form.getValue('assigned_to') == "") {
g_form.showErrorMessage('assigned_to', 'Assigned To is mandatory for High impact incidents.');
```

onCellEdit Client Script - Prevent state change via list edit:

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) { alert('State cannot be updated using list
editing. Please open the Incident.');
```

Dataset Link

Not applicable - configuration-based project on ServiceNow platform.

GitHub & Project Demo Link

GitHub: <https://github.com/raman2662000-cpu/Implement-Client-Script-UI-Policy-Incident-.git>

Project Demo Video: <https://youtu.be/EEuYLnaIMII>